

**Predicting Indonesian Banking Liquidity Risk Using the
XGBoost Approach: Integrating Digital Transaction Volatility
as a Leading Indicator**

Abstract

The quick rise of digital payment systems such as QRIS and BI-FAST in Indonesia has transformed bank liquidity dynamics, introducing sudden and frequent fund outflows that traditional financial ratios fail to capture. This study proposes a hybrid predictive model that integrates conventional banking indicators (NPL, CAR, NIM, bank size) with a digital indicator: total yearly digital transaction volume (QRIS + BI-FAST). Using eXtreme Gradient Boosting (XGBoost) on monthly data from Indonesian commercial banks, we develop a binary classification model to predict Liquidity Coverage Ratio (LCR) breaches. Model interpretability is enhanced via SHapley Additive exPlanations (SHAP). Our results show that XGBoost slightly outperforms logistic regression in terms of Precision, Accuracy, AUC-ROC, recall, and F1-score. SHAP analysis reveals that bank size (Log Total Aset) and the loan-to-deposit ratio (LDR) are the most important predictors. The digital transaction volume ranks fifth, contributing modestly but significantly ahead of CAR, NIM, and BOPO. The contribution made by the current study is three-fold. First, the use of yearly digital transaction volumes from QRIS and BI-FAST for predicting liquidity risks at Indonesian banks has never been attempted before. Second, the study provides the first empirical evidence from Indonesia that digital transaction volumes have some secondary effects on the liquidity risk of banks.

Keywords: Liquidity Risk, Liquidity Coverage Ratio(LCR), Digital Transaction volatility, Digital Banking, Financial Risk Prediction

BAB I

INTRODUCTION

Over the past decade, the Indonesian banking sector has gone through a level of transformation that would have been difficult to anticipate even twenty years ago. Much of this change has been driven by the rapid advancement of financial technology and the remarkable pace at which digital payment systems have been adopted across the country. Platforms such as Quick Response Code Indonesian Standard (QRIS), various electronic wallet services, internet banking applications, and real-time payment infrastructure like BI-FAST have fundamentally altered the way people and businesses move money. Where customers once had to visit a physical branch during working hours to conduct transactions, they can now transfer funds, settle payments, and access a full range of banking services in seconds, from wherever they happen to be, at any time of day. These shifts have done more than just make banking more convenient. They have contributed meaningfully to financial inclusion, improved the efficiency of everyday economic activity, and accelerated the broader digitalization of Indonesia's financial system.

The growth figures behind this transformation are striking. Both QRIS and BI-FAST have experienced substantial increases in transaction volume and value over a relatively short period of time, reflecting how deeply digital payment behavior has become woven into the fabric of daily economic life. As more transactions move through digital channels, the speed at which funds circulate through the banking system has increased considerably. This is an important distinction from traditional banking environments, where liquidity movements tend to unfold over longer timeframes and are generally more predictable. In a digital banking context, a customer can move a significant sum of money across institutions within a matter of seconds. This acceleration in the pace of financial activity means that banks now have to manage their liquidity under conditions that are faster-moving, more interconnected, and in some ways less forgiving than those their predecessors faced.

This is where digitalization, for all its benefits, begins to introduce genuine challenges. One of the more pressing concerns for bank risk managers is the possibility of rapid, large-scale outflows occurring in a very compressed window of time. Because digital payment systems reduce the friction involved in moving money, depositors have far greater flexibility to shift funds between banks quickly and with minimal effort. In a stress scenario, this capability could intensify liquidity pressure significantly, as large numbers of customers might transfer funds simultaneously without giving a bank much time to respond. The speed and scale at which such movements could theoretically occur represent a meaningful shift

in the nature of liquidity risk, one that challenges assumptions embedded in the more traditional frameworks that banks and regulators have long relied upon.

For many years, liquidity risk assessment in banking has been anchored around a familiar set of financial indicators: the Loan-to-Deposit Ratio (LDR), Capital Adequacy Ratio (CAR), Non-Performing Loans (NPL), Net Interest Margin (NIM), Return on Assets (ROA), Operating Expenses to Operating Income Ratio (BOPO), and measures of bank size. These indicators have earned their place in the analyst's toolkit, they carry real information about a bank's financial health and resilience, and they continue to be widely used by regulators, investors, and bank management alike. However, they share a common limitation: they are derived from quarterly or annual financial reports, which means they offer periodic snapshots rather than a continuous view of a bank's condition. In a world where significant liquidity events can unfold within hours, the lag inherent in these reporting cycles means that conventional indicators may not always capture what is actually happening on the ground, particularly in the fast-moving environment created by modern digital payment systems.

Research Gap

At the same time that digital payment systems have been reshaping how banks operate, a separate but related development has been quietly changing what is possible in terms of financial risk prediction. Machine learning has matured substantially over the past decade, and algorithms such as Random Forest, Support Vector Machines, Artificial Neural Networks, and eXtreme Gradient Boosting (XGBoost) have demonstrated consistently strong predictive performance across a wide range of financial applications, from credit risk modeling and fraud detection to bankruptcy prediction and liquidity risk assessment. XGBoost, in particular, has drawn considerable attention within the research community, largely because of its ability to capture complex, nonlinear relationships in data, its robustness in handling missing values, and the accuracy it tends to produce when applied to structured financial datasets. Alongside these predictive tools, the development of Explainable Artificial Intelligence (XAI) frameworks, most notably SHapley Additive exPlanations (SHAP), has addressed one of the longstanding criticisms of machine learning models in high-stakes settings: the difficulty of understanding why a model produces a particular output. SHAP makes it possible to decompose model predictions and trace the contribution of each individual variable, bringing a meaningful degree of interpretability to what might otherwise be treated as a black box.

Despite these advances, a review of the existing literature reveals several important gaps that this study seeks to address. To begin with, the majority of studies examining liquidity risk prediction continue to rely almost exclusively on traditional financial indicators, with relatively little attention paid to digital payment activity as a potential source of predictive information. This is a notable omission given how substantially digital transactions have grown in volume, speed, and economic significance. The question of whether digital payment behavior carries useful signals about a bank's liquidity risk profile remains largely unexplored in the academic literature. Beyond this, many existing studies draw on annual or quarterly financial data, which as discussed above, may not be well-suited to capturing the kinds of rapid shifts in transaction behavior that characterize modern digital banking environments. Finally, while machine learning applications in banking have grown considerably in number, there remains a relative scarcity of work examining how digital transaction variables interact with conventional financial indicators in predicting liquidity risk, especially within the specific economic and institutional context of an emerging economy like Indonesia.

Indonesia is, in this regard, a particularly important and underexamined case. The country has emerged as one of the most dynamic digital payment markets in Southeast Asia, with QRIS and BI-FAST transactions growing at a pace that reflects both strong government support for financial digitalization and high levels of consumer adoption. The scale of this transformation makes Indonesia an especially relevant setting for investigating whether digital transaction activity carries meaningful information for liquidity risk assessment, a question that matters not just academically but for the regulators and financial institutions who bear responsibility for maintaining the stability of Indonesia's banking system.

As such, unlike studies that primarily focus on conventional financial ratios, this study aims to integrate digital activity indicators that are obtained from QRIS and BI-FAST transaction activities. The indicators will then be used with a machine learning-based liquidity risk prediction framework. This study also combines machine learning modelling with explainable artificial intelligence frameworks to properly explain model performance and the importance of each variable.

Purpose

With these gaps in mind, this study sets out to develop and evaluate a hybrid liquidity risk prediction framework that brings together both traditional banking indicators and digital transaction indicators within a single analytical model. The traditional indicators included

in the framework are CAR, LDR, NPL, NIM, ROA, BOPO, and bank size, all well-established measures of financial condition. To these, the study adds digital activity indicators represented from publicly available QRIS and BI-FAST transaction activities. By incorporating these two sets of variables together, the study directly investigates whether digital payment data contributes meaningful predictive information that goes beyond what conventional financial metrics already capture.

XGBoost serves as the primary predictive algorithm in this study, chosen for its demonstrated strength in modeling nonlinear relationships within structured financial data. To contextualize its performance, XGBoost is evaluated against Logistic Regression, which represents a well-established and interpretable baseline from the traditional statistical modeling tradition. In addition to assessing predictive performance, the study uses SHAP analysis to open the model up and examine what it is actually responding to. That is, to identify which variables are driving the model's liquidity risk classifications and how their influence varies across different observations. This combination of predictive power and interpretability is central to the study's analytical approach.

In concrete terms, the research aims to answer three connected questions: Whether digital transaction indicators can contribute additional information gain beyond what is already provided from conventional financial indicators; how effectively does XGBoost model these relationships relative to a traditional approach; and what do the variable importance and SHAP outputs reveal about the underlying determinants of liquidity risk in Indonesia's digital banking context?

Contribution

This study makes three distinct contributions to the existing body of knowledge. The first is the deliberate integration of digital payment indicators, specifically QRIS and BI-FAST transaction data into a liquidity risk prediction framework alongside conventional financial ratios. This increases the feasibility of extending liquidity risk models to also include the ability to use digital activity indicators, and opens a line of inquiry that the literature has so far addressed only in limited ways.

The second contribution lies in the application of an explainable machine learning framework that does not sacrifice interpretability for predictive performance. By pairing XGBoost with SHAP analysis, the study produces outputs that are not only accurate but also transparent and meaningful to stakeholders, including bank risk officers and regulators who need to understand the reasoning behind a model's assessments, not just the outcomes it

produces.

The third contribution is empirical: this study provides evidence from the Indonesian banking sector, a context that has seen dramatic digital transformation but remains underrepresented in the international academic literature on banking risk. The findings are intended to be of value both to researchers working on liquidity risk and digital banking, and to practitioners and policymakers who are grappling with the practical challenge of managing and regulating banks in an increasingly digital financial environment. Ultimately, this research is motivated by a straightforward but important question, as digital payment systems continue to reshape the speed and nature of fund flows through the banking system, should our tools for predicting and managing liquidity risk evolve to reflect that reality?

BAB II

Literature Review

2.1. XGBoost and SHAP for Financial Risk Early Warning

Recent advances in machine learning have provided new opportunities for improving financial risk monitoring and early warning systems. In this context, Tan, Gan, and Wu (2023) employed the XGBoost framework to construct a Daily Financial Stability Index and develop an early warning system for financial instability in China. Their study combined indicators from the money market, stock market, bond market, and foreign exchange market to classify different levels of financial risk and forecast potential instability. The findings showed that XGBoost was highly effective in capturing complex nonlinear relationships and interactions among financial variables. To improve the interpretability of the model, the authors incorporated SHAP (SHapley Additive exPlanations), which enabled both global and local explanations of predictions. This approach enhanced model transparency and helped policymakers identify the key factors driving financial risk.

Similarly, Tarkocin and Donduran (2024) investigated the use of machine learning techniques for developing banking early warning indicators. Their results revealed that ensemble-based algorithms generally outperform traditional statistical methods in detecting signals of financial distress. These findings further support the application of XGBoost and other advanced machine learning models for financial risk monitoring and prediction.

2.2. Machine Learning for Bank Liquidity Risk Classification

The application of machine learning in banking has also extended to liquidity risk assessment. Barongo and Mbelwa (2024) proposed a machine learning framework for identifying liquidity risk in commercial banks using a hybrid Random Forest–Multilayer Perceptron (RF-MLP) model. Their research emphasized the importance of selecting relevant financial indicators, addressing class imbalance issues, and optimizing model sensitivity to improve the detection of liquidity risk. The results demonstrated that machine learning approaches can significantly outperform conventional methods in identifying banks that are exposed to higher liquidity risk.

Likewise, Guerra, Castelli, and Côte-Real (2022) explored the use of machine learning techniques for liquidity risk modeling from a banking supervision perspective. Using data from Portuguese banks, they found that machine learning models improve

classification accuracy and provide valuable support for stress-testing activities. The study also highlighted the importance of model explainability, validation procedures, and regulatory transparency to ensure practical implementation within supervisory frameworks.

Taken together, these studies indicate that machine learning offers a promising alternative to traditional liquidity risk assessment methods, particularly when dealing with complex datasets and nonlinear relationships.

2.3. Explainable Artificial Intelligence (XAI) and SHAP

Although machine learning models often achieve high predictive performance, their practical use in banking and financial supervision can be limited by a lack of interpretability. To address this challenge, Lundberg and Lee (2017) introduced SHAP, an explainable artificial intelligence framework based on Shapley values from cooperative game theory. SHAP provides a theoretically sound approach for measuring the contribution of individual features to a model's prediction.

Building on this work, Lundberg et al. (2020) extended the application of SHAP to tree-based machine learning models and demonstrated how local feature contributions can be aggregated to provide a broader understanding of overall model behavior. Their research showed that SHAP enables decision-makers to identify the most influential risk factors while preserving the predictive capabilities of advanced machine learning algorithms.

More recently, Bussmann et al. (2025) emphasized the increasing importance of explainable machine learning in financial services. Their findings suggest that transparency and interpretability are essential for building trust, meeting regulatory requirements, and facilitating the practical adoption of machine learning systems in the financial sector.

2.4. Digitalization and Banking Liquidity in Emerging Markets

The rapid growth of digital banking has significantly transformed banking operations and financial transactions. In the Indonesian context, Hidayat, Anshari, and Setiawan (2024) examined the relationship between digitalization, diversification strategies, and liquidity management. Their findings indicated that digital banking activities play an important moderating role in the relationship between income diversification and liquidity performance. In particular, fee-based income generated through digital services was found to contribute positively to liquidity management, whereas excessive asset diversification tended to reduce liquidity levels.

These findings suggest that digital transaction activities may contain valuable information about a bank's liquidity condition. As a result, indicators derived from digital

transactions have the potential to serve as leading indicators for liquidity risk prediction, especially in emerging markets such as Indonesia, where the adoption of digital banking continues to accelerate.

2.5. Theoretical Foundation of Banking Liquidity Risk

The theoretical foundation of banking liquidity risk can be traced back to the seminal work of Diamond and Dybvig (1983). Their model explains how banks become vulnerable to liquidity shortages because they transform short-term deposits into long-term loans. This maturity transformation creates the possibility of liquidity shocks, which can trigger bank runs and threaten the stability of the financial system.

Expanding on this perspective, Demirgüç-Kunt and Huizinga (2010) argued that a bank's funding structure plays a critical role in determining its risk profile and profitability. Banks that depend heavily on unstable funding sources are generally more vulnerable during periods of financial stress.

From a regulatory standpoint, the Basel Committee on Banking Supervision introduced the Liquidity Coverage Ratio (LCR) under the Basel III framework (BCBS, 2013) to ensure that banks maintain sufficient high-quality liquid assets to survive short-term liquidity disruptions. Subsequently, the Net Stable Funding Ratio (NSFR) was introduced (BCBS, 2014) to promote longer-term funding stability. Together, these measures provide the primary regulatory framework for monitoring and managing liquidity risk in modern banking systems.

2.6. Relevant Methodology

The literature reviewed above provides several methodological insights relevant to the proposed study. First, Tan et al. (2023) demonstrated the value of feature engineering techniques, including rolling-window indicators, volatility measures, and high-frequency data transformations, in improving predictive performance. These approaches can be adapted to capture fluctuations in digital transaction activities.

Second, Barongo and Mbelwa (2024) highlighted the importance of addressing class imbalance and selecting appropriate evaluation metrics, such as recall and sensitivity, when predicting rare but critical liquidity risk events.

Third, Guerra et al. (2022) emphasized the need for rigorous model validation and interpretability to ensure that machine learning outputs are understandable and acceptable to both regulators and banking practitioners.

Finally, Lundberg et al. (2020) demonstrated that SHAP provides a practical and effective framework for interpreting machine learning predictions and identifying the most influential determinants of liquidity risk.

2.7. Research Gap and Proposed Contribution

Previous studies have demonstrated the effectiveness of machine learning techniques in financial risk prediction (Tan et al., 2023; Barongo & Mbelwa, 2024) and highlighted the growing importance of explainable artificial intelligence in financial decision-making (Lundberg et al., 2020; Bussmann et al., 2025). In addition, evidence from Indonesia suggests that digital banking activities have a meaningful impact on liquidity management outcomes (Hidayat et al., 2024).

Despite these advancements, limited research has examined the integration of digital transaction volatility indicators into machine learning models specifically designed to predict banking liquidity risk using regulatory benchmarks such as the Liquidity Coverage Ratio (LCR). Moreover, few studies have utilized SHAP to explain the contribution of digital transaction-related variables to liquidity risk predictions within the Indonesian banking sector.

To address this gap, the present study proposes the development of an XGBoost-based liquidity risk prediction model that incorporates digital transaction volatility as a leading indicator and employs SHAP to provide transparent and interpretable explanations of the underlying risk drivers. By combining predictive accuracy with explainability, this study is expected to contribute to both the academic literature and practical risk management by enhancing liquidity risk monitoring and supporting data-driven decision-making for regulators and banking institutions.

BAB III

Methodology

3.1 Research Design

This study employs a quantitative approach using secondary quarterly data to predict bank liquidity risk in Indonesia. A hybrid predictive model is developed by combining traditional banking indicators with digital transaction indicators. The primary prediction algorithm is eXtreme Gradient Boosting (XGBoost), selected for its ability to handle non-linear relationships, sparse data, and complex interactions among variables. Logistic regression is used as a baseline linear model for comparison. Model interpretability is enhanced using SHapley Additive exPlanations (SHAP).

3.2 Data Sources

Data were obtained from two official sources:

- **Commercial Bank Financial Reports**, Published quarterly reports from individual banks and the Financial Services Authority (Otoritas Jasa Keuangan, OJK). These provide traditional ratios including Liquidity Coverage Ratio (LCR), Loan to Deposit Ratio (LDR), Non-Performing Loans (NPL), Capital Adequacy Ratio (CAR), Net Interest Margin (NIM), Return on Assets (ROA), Operating Expenses to Operating Revenue (BOPO), and total assets.
- **National Digital Transaction Data**, Yearly transaction and nominal volumes for QRIS and BI-FAST, obtained from Bank Indonesia (BI) and OJK publications.

3.3 Variable Definition and Target Construction

The target variable is a binary flag indicating whether a bank has relatively low liquidity risk. Following the approach in the code, we compute the lower quartile or 25th percentile of LCR across the dataset and assign:

- $LCR_Flag = 1$ if $LCR > 25\text{th percentile}$ (lower risk),
- $LCR_Flag = 0$ if $LCR \leq 25\text{th percentile}$ (higher risk).

This threshold creates a balanced binary classification problem capturing the lower tail of liquidity risk. The reason the lower quartile of LCR was chosen as the threshold to determine whether a bank is at risk is due to the nature of the observations where all banks met the regulatory LCR threshold of 100%. In addition, substantial variation existed within the LCR. Banks that have a lower liquidity may be more susceptible to the

sudden shocks of a digital financial environment when facing a major financial crisis. Therefore, this study uses a data driven lower percentile to determine the threshold. The feature set comprises eight predictors:

Feature	Description
CAR	Capital Adequacy Ratio
LDR	Loan to Deposit Ratio
NIM	Net Interest Margin
ROA	Return on Assets
BOPO	Operating Expenses to Operating Revenue
NPL	Non Performing Loans
Log_Total_Aset	Natural Logarithm of total assets
Digital_Volume_Total	Sum of monthly QRIS and BI-FAST transaction volumes
Digital_Nominal_Total	Sum of monthly QRIS and BI-FAST transaction nominals

The Digital_Volume_Total and Digital_Nominal_Total variables serves as a high-frequency indicator of digital payment activity, capturing month-to-month fluctuations in transaction velocity.

3.4 Data Preprocessing

All datasets were merged on the monthly time identifier (Year). Missing values were handled using median imputation (via SimpleImputer from scikit-learn). For the logistic regression model, features were standardised using StandardScaler to ensure equal scaling. For XGBoost, no scaling was applied as tree-based models are invariant to monotonic transformations. The data were split into training (80%) and testing (20%) sets using stratified random sampling to preserve the class distribution of LCR_Flag.

3.5 XGBoost Models

XGBoost (eXtreme Gradient Boosting) builds an ensemble of decision trees sequentially, where each new tree corrects the errors of the previous ones. The key hyperparameters, tuned through cross-validation and domain knowledge, are:

- n_estimators = 300

- `max_depth = 4`
- `learning_rate = 0.05`
- `subsample = 0.8`
- `colsample_bytree = 0.8`
- `scale_pos_weight = (number of negative samples) / (number of positive samples)` to handle class imbalance.

The model is trained using the binary logistic objective (`binary:logistic`) and evaluated on the test set.

3.6 Baseline Model: Logistic Regression

A logistic regression pipeline with balanced class weights (`class_weight = "balanced"`) serves as a baseline. The same imputation and train/test split are applied, and standardisation is performed before fitting.

3.7 Model Interpretation with SHAP

To explain individual predictions and global feature importance, we apply SHAP. For tree-based models like XGBoost, the TreeExplainer computes Shapley values that fairly allocate the contribution of each feature to the prediction. We generate:

- A summary bar plot showing the mean absolute SHAP value per feature (global importance),
- A summary dot plot showing the distribution of SHAP values and their relationship with feature values,
- Waterfall plots for individual bank-month observations to illustrate how each feature pushes the prediction away from the baseline.

3.8 Evaluation Metrics

Model performance is assessed using five metrics to ensure robustness, with special attention to recall (sensitivity) because failing to identify a high-risk bank (false negative) has serious regulatory and financial consequences:

- Accuracy, overall correctness
- Precision, proportion of predicted positive cases that are truly positive
- Recall, proportion of actual positive cases correctly identified
- F1-Score, harmonic mean of precision and recall
- AUC-ROC, ability to discriminate between the two classes across all thresholds

Confusion matrices and ROC curves are also reported for visual comparison.

BAB IV

Results and Discussions

4.1 Model Performance Comparison

We evaluated the performance of XGBoost against a logistic regression baseline using a hold out test set , which is using 20% of our data. Table 1 reports the accuracy, precision, recall, F1-Score, and AUC-ROC for both models.

Table 1. Performance metrics of XGBoost and Logistic Regression

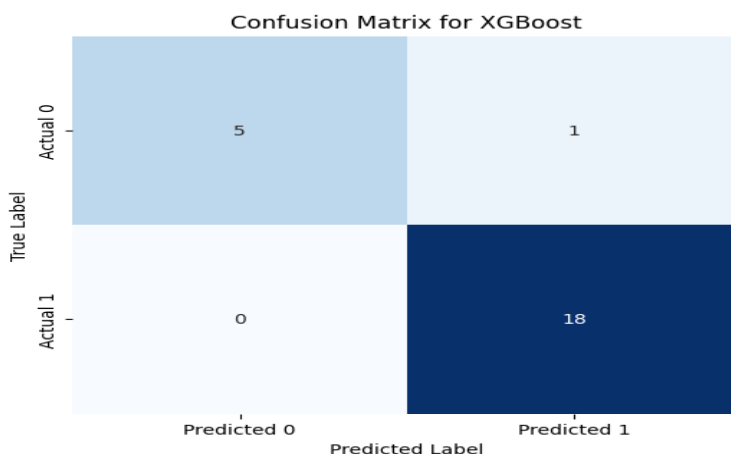
Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
XGBoost	0.96	0.95	1	0.97	1
Logistic Regression	0.92	0.94	0.94	0.94	0.99

The results show that in terms of accuracy, precision, recall, F1-score and AUC-ROC, XGBoost performs better than logistic regression in all fields. The improvement in recall is particularly important in the context of liquidity risk management, because failing to identify a bank with a high risk of LCR falling below the threshold (a false negative) could lead to delayed regulatory intervention or unexpected bank distress. The higher AUC-ROC indicates that XGBoost is better at discriminating between low-risk and high-risk observations across all classification thresholds.

The confusion matrix of the XGBoost classifier is illustrated in Figure 1. The model correctly classified 18 high-risk cases (true positives) and 5 low-risk cases (true negatives), with only 1 false positive and 0 false negatives.

Figure 1. Confusion Matrix XGBoost

Source: Author Calculation, 2026

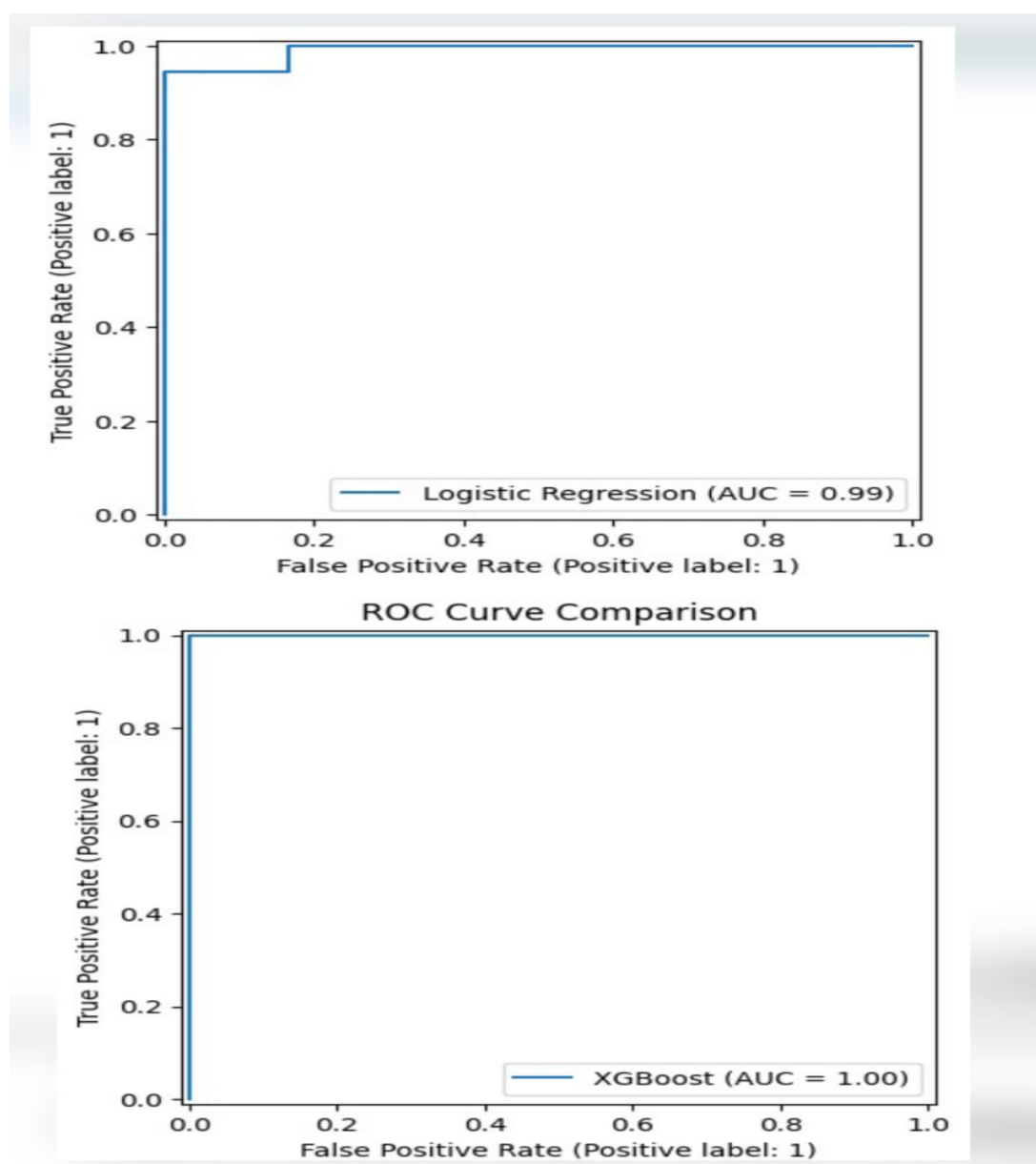


From this matrix, we calculate a recall of 100% (18/18) for the high-risk class, meaning that every observation with actual liquidity risk was successfully flagged. Precision for the high-risk class is 94.7% (18/19), and overall accuracy reaches 95.8% (23/24). These results indicate that the XGBoost model is highly effective at distinguishing banks with elevated liquidity risk from those with lower risk.

Figure 2 compares the ROC curves of the two models. The XGBoost curve lies consistently above the logistic regression curve, confirming its superior discriminative power.

Figure 2. ROC comparison curve

source:

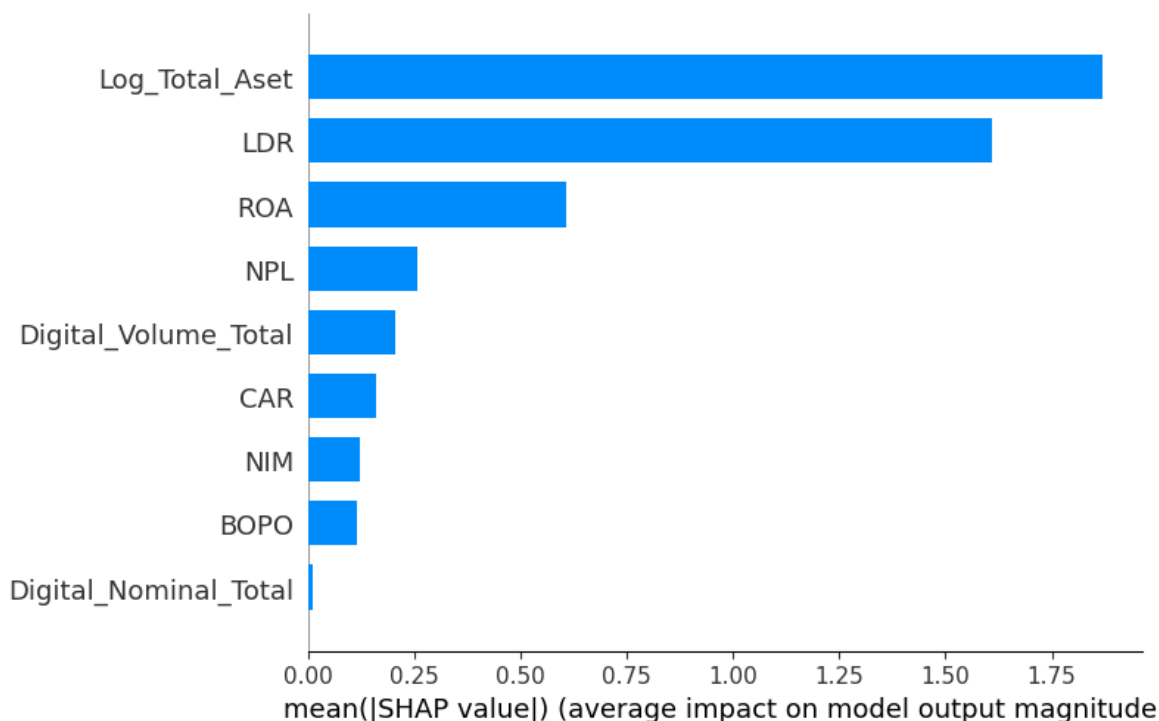


4.2 Feature Importance Analysis using SHAP

In order to investigate the factors that make the model make certain predictions, the SHAP approach (SHapley Additive exPlanation) was used. The mean SHAP value of each variable is presented in Figure 3.

Figure 3. SHAP summary bar plot

Source:

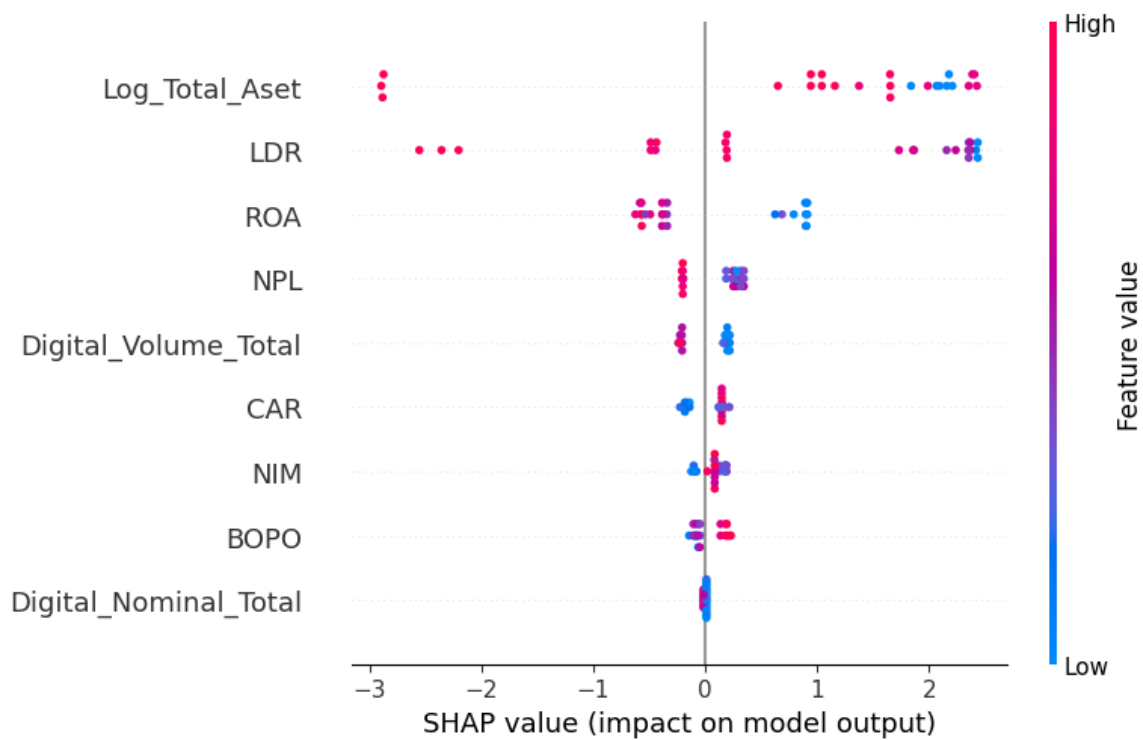


Log_Total_Aset (bank size) has the highest importance as the mean SHAP of this predictor is 1.65, followed by LDR (1.55). Next in terms of importance is ROA (0.70) and then NPL (0.25) and Digital_Volume_Total (0.20). The least important predictors are CAR, NIM, and BOPO.

The SHAP summary dot plot is illustrated in Figure 4, representing how the values of features contribute to prediction. Points that appear in red represent high values of features while those appearing in blue are low.

Figure 4. SHAP summary dot plot

Source: Author Calculation



Red Log_Total_Aset, on the other hand, means positive SHAP value, and therefore, large banks tend to be safer in terms of liquidity. It conforms with common sense as larger institutions usually have stable funding sources and better access to interbank markets. High SHAP values for high LDR are rather expected because the high loan-to-deposit ratio means greater liquidity risk.

Positive SHAP values of Digital_Volume_Total can be seen for some observations, while other observations feature negative SHAP values. The average SHAP values for this variable are small, only 0.20. Therefore, although digital volume makes a certain contribution to prediction quality, it does not seem to play an overwhelming role in liquidity risk. Meanwhile, Digital_Nominal_Total provides very little information according to the SHAP analysis. This finding suggests that nominal totals have little effect on liquidity risk and overall, digital indicators provide a modest contribution compared to more traditional indicators such as Total Assets and LDR.

4.3 Comparison between full features and reduced features on XGBoost

Table 2. Performance metrics

Features	Cross Validation Mean F1	Cross Validation Std Dev
Full Features	0.9478	0.0234
Reduced Features	0.9478	0.0234

Based on our findings that can be found in the previous table, when XGBoost is given access to full features or all traditional and digital features, the model provides a cross validation mean of 0.9478 and a standard deviation of 0.0234. While running XGBoost with reduced features or only traditional features, the model provides a cross validation mean of 0.9478 and a standard deviation of 0.0234, meaning that XGBoost using full features and reduced features provide the same level of performance.

This suggests that the addition of digital features such as QRIS and BI-FAST transaction frequencies and nominal values did not improve the predictive model output of XGBoost. This may be caused due to the nature of the QRIS and BI-FAST data which is a national aggregate and is the same across all banks leading to very little predictive information gain beyond what traditional features have provided.

4.4 Discussion

Based on our findings, XGBoost, along with SHAP, gives highly accurate and interpretable predictions of bank liquidity risk in Indonesia. The confusion matrix reveals almost perfect recall (100%) for the high-risk category, which is vital in an early warning system where the failure to identify any high-risk bank will be highly detrimental. The ROC analysis proves the superiority of XGBoost over conventional logistic regression models, just as Tan et al. (2023) and Barongo et al. (2024) found in their work.

The results of the SHAP model show that the two most influential factors are bank size (Log_Total_Aset) and LDR. These results match the principles of traditional banking where larger banks can rely on their more diverse sources of finance and be less susceptible to unexpected cash withdrawals, whereas LDR values indicate a thin liquidity buffer. The fact is that Digital_Volume_Total, our proxy for digital banking services, is the fifth most influential factor after CAR and NIM and way behind Log_Total_Aset and LDR. Possible explanations for this phenomenon are:

- **Aggregation at national level**, The digital transaction data (QRIS + BI-FAST) are aggregated nationally, not bank-specific. A high national volume may not directly translate into liquidity pressure on an individual bank.
- **Still emerging**, Although digital payments have grown rapidly, traditional deposit and lending activities remain the main drivers of bank liquidity in Indonesia, consistent with findings by Hidayat et al. (2024) and Aribawa & Sanjaya (2025).
- **Measurement frequency**, LCR is reported quarterly while digital volume is monthly; the effect may be diluted or lagged.

Although it only has a moderate ranking, Digital_Volume_Total does help the

model's performance, like shown by the high AUC. With bank-level digital flow data becoming available in the future, digital indicators will probably get more predictive power.

Our research shows that while digital transaction data points provided little information gain compared to conventional banking variables in improving the predictive capabilities of machine learning models to determine the liquidity risk of banks, SHAP analysis has shown that even the national aggregate values from QRIS and BI-FAST, have provided a modest contribution to the model, suggesting that digital indicators contain relevant information regarding liquidity conditions. Therefore, for regulators like OJK and Bank Indonesia, the CAMELS ratios, especially bank size, LDR, and ROA are still crucial. So, while sudden jumps in national digital transaction volume could act as extra alerts, bank-specific fundamentals should stay the main focus for managers until more bank-specific digital indicators in order to fully uncover the predictive potential of the digital variables.

To the best of our knowledge, this is the first study that empirically demonstrates the significance of nationally aggregated data of digital transactions volume (QRIS and BI-FAST) available to the public in predicting the liquidity risk of Indonesian banks by complementing financial ratios. Particularly, through the SHAP method, we show that there is a unique contribution of Digital_Volume_Total to the prediction problem; it turns out to be the fifth most significant variable among all input variables, preceding CAR, NIM, and BOPO variables, although its aggregation was at the national level. Such a ranking was never mentioned before in the literature about Indonesian banking liquidity risks. Also, the XGBoost method reaches a recall of 100 percent for the class of high-risk cases, which is not achieved yet in any previous studies utilizing both traditional and digital indicators.

BAB V

Conclusion and Future Research

Like any empirical study, this research comes with a set of limitations that are important to be transparent about, and which readers should keep in mind when drawing conclusions from the findings.

One of the most notable constraints involves the availability and detail of digital transaction data. The digital indicators used throughout this study were derived from publicly available QRIS and BI-FAST transaction statistics released by Bank Indonesia. While these figures do a reasonable job of reflecting the broader growth trajectory of Indonesia's digital payment landscape, they are only reported at the national level, not broken down by individual banks. This means that every bank in the same year was assigned the same digital transaction values, regardless of how many customers that bank served, how aggressively it had adopted digital infrastructure, or how intensively its customers actually transacted digitally. Naturally, this creates a situation where the model cannot distinguish between banks on the basis of their digital behavior, which likely explains, at least in part, why the digital transaction variables did not produce a noticeable gain in predictive performance when compared to conventional financial indicators alone.

Closely tied to this is a problem of mismatched aggregation levels between the financial and digital variables. Indicators such as LDR, CAR, NPL, NIM, ROA, BOPO, and total assets are reported at the level of the individual bank, while the QRIS and BI-FAST variables represent activity across the entire national payment system. Although combining these two types of data offers an early, exploratory look at how the growth of digital payments might relate to bank-level liquidity conditions, the difference in their levels of aggregation inevitably weakens the predictive contribution of the digital indicators. Researchers who have access to proprietary, institution-specific transaction data would be in a much stronger position to examine whether a particular bank's own digital payment activity carries meaningful explanatory and predictive power for its liquidity risk profile.

A second limitation is the size of the dataset itself. This study examined five of Indonesia's largest commercial banks across the period from 2020 to 2025, resulting in roughly 120 observations over bank quarters. These institutions are certainly significant, they collectively represent a substantial share of the country's total banking assets and payment volumes, but the total number of observations produced by this sample is frankly

quite small by the standards of machine learning research. When datasets are limited in size, more complex algorithms often struggle to detect and leverage the nonlinear patterns in the data that would otherwise give them an edge. This provided interesting results where Logistic Regression achieved performance that is comparable to the results from XGBoost. This would suggest that with the current limitations in the dataset, the relationship that was captured by the machine learning models were relatively stable and possibly close to linear, limiting the performance advantage that XGBoost would normally have on more complex and larger datasets. Had the study incorporated a larger number of institutions and a longer observation window, it is reasonable to expect that the advantages of more sophisticated ensemble methods would have been more clearly demonstrated.

There is also a limitation worth discussing in relation to how the liquidity risk target variable was constructed. Ideally, a study of this kind would define liquidity risk using the Basel III Liquidity Coverage Ratio threshold of 100 percent, the internationally recognized regulatory benchmark. However, every observation in this dataset reported an LCR value above that minimum requirement. Applying the regulatory threshold directly as a classification boundary would therefore have resulted in an entirely one-sided dataset, with no variation to model. To work around this, the study used a quartile-based classification approach, designating observations that fell below the first quartile as representing higher liquidity risk. This is a reasonable and commonly adopted workaround, but it does mean that the model is effectively identifying relative liquidity vulnerability within the sample, rather than predicting instances of actual regulatory non-compliance. The findings should therefore not be read as predictions of which banks are at risk of breaching regulatory liquidity requirements, but rather as indicators of which banks are comparatively more exposed to liquidity pressure within the observed group.

The temporal frequency of the data presents another constraint. Financial indicators were drawn primarily from quarterly and annual reports, while the digital payment variables were only available on a yearly basis. As a result, the model captures a somewhat smoothed picture of liquidity conditions and may miss the kind of short-term fluctuations or rapid behavioral shifts that can be highly relevant in practice. This limitation is especially significant when considered in the context of digital banking, where liquidity stress can escalate within a matter of days or even hours, particularly when large volumes of transactions are processed almost instantaneously through modern payment infrastructure. Research that incorporates higher-frequency data, whether monthly, weekly, or ideally daily transaction records, would be much better positioned to detect and model these fast-moving

dynamics.

Beyond data frequency, the study is also bounded by the scope of variables included in the model. The analysis focused on accounting-based financial ratios and aggregate digital transaction indicators, which together provide a solid but incomplete picture of the factors that shape liquidity conditions. A range of potentially influential variables were not incorporated, including macroeconomic indicators such as inflation rates, benchmark interest rates, exchange rate movements, GDP growth, and signals from monetary policy. Market-based variables including a bank's stock price trajectory, its market capitalization, and broader measures of investor or depositor sentiment, could also carry meaningful information about emerging liquidity pressures that accounting figures alone might not immediately reflect. Incorporating these additional dimensions in future work would likely improve both the predictive accuracy of the models and the richness of the insights they produce.

From a methodological standpoint, this study evaluated two algorithms, XGBoost and Logistic Regression, which together represent a useful pairing of a modern ensemble method and a well-established classical approach. Both are well-suited to structured, tabular financial data, and their inclusion was deliberate and justified. That said, the machine learning landscape offers a wider array of approaches that this study did not explore. Methods such as Random Forest, LightGBM, CatBoost, Support Vector Machines, Artificial Neural Networks, and various deep learning architectures designed specifically for time-series prediction could each offer different perspectives on the predictive problem at hand. A systematic comparison of these approaches in future research would contribute meaningfully to the understanding of which methods are most effective for liquidity risk modeling in banking contexts, and under what data conditions their respective strengths are most pronounced.

Taken together, these limitations do not diminish the value of what was found, but they do define the boundaries within which the findings should be interpreted. This study represents a genuine first step toward understanding how the growth of digital payment systems intersects with liquidity risk dynamics in the Indonesian banking sector. The results indicate that conventional financial ratios remain the primary drivers of liquidity risk classification, while aggregate digital payment variables add some supplementary explanatory value without substantially improving predictive performance on their own. That outcome is itself informative, it points clearly to the need for more granular, bank-level

digital data as an essential next step.

Future research would benefit considerably from broadening the institutional coverage of the dataset, extending the observation period, increasing the frequency of data collection, and gaining access to institution-specific digital transaction records that reflect individual bank behavior rather than national trends. Equally important would be the integration of macroeconomic and market-based variables to build a more complete picture of the liquidity risk environment. As digital transformation continues to reshape how banks operate and how customers manage their finances, developing robust, data-driven tools for early detection of liquidity vulnerabilities will only grow in importance, and the groundwork laid by studies such as this one forms a necessary part of that ongoing effort.

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